
From Noise to Diversity: Random Embedding Injection in LLM Reasoning

Anonymous Author(s)

Affiliation

Address

email

Abstract

Recent *soft prompt* research has tried to improve reasoning by inserting trained vectors into LLM inputs, yet whether the gain comes from the *learned content* or from the *act of injection itself* has not been carefully separated. We study Random Soft Prompts (RSPs), which drop the training step entirely and append a freshly drawn sequence of random embedding vectors to the input. Each RSP vector is sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table; the sequence carries no learned content, and yet reaches accuracy comparable to optimized soft prompts on math reasoning benchmarks in several settings. The mechanism unfolds in two stages: because attention has to absorb a never-seen-before random position, the distribution over the first few generated tokens flattens and reasoning trajectories *branch*, and as generation continues this influence dilutes naturally so the response commits to a single completion. We show that during inference RSPs lift early-stage token diversity and, combined with temperature sampling, widen Pass@ N , the probability that at least one out of N attempts is correct. Beyond inference, we carry the same effect into DAPO training and demonstrate practical gains. Our contributions are: (i) RSP isolates the simplest form of soft prompt — training-free, freshly resampled — providing a unified lens for the structural effect of injection that variants otherwise differing in training and form all share; (ii) an empirical and theoretical validation of the underlying mechanism; and (iii) an extension from inference to training.

1 Introduction

This work began from a surprising observation: even *untrained*, randomly drawn embeddings appended to LLM inputs lift reasoning accuracy and reshape early decoding in several settings — with no optimization at all. As LLMs scale, full fine-tuning is prohibitive, motivating parameter-efficient methods [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* insert learnable continuous embeddings into the input and steer behavior through attention [Li and Liang, 2021, Lester et al., 2021]; the paradigm is actively adopted for mathematical reasoning [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]. These methods diverge in injection position, training objective, and prompt form, and prior work attributes their gains to the *learned content* of the optimized vectors. What every variant shares — the act of injecting embeddings into the input — has remained outside the analysis.

To separate *learned content* from the *act of injection*, we use random soft prompts (RSPs) as a control — continuous vectors drawn from an isotropic Gaussian fitted to the embedding table’s mean and variance, with no learning. The control turns out to be non-neutral. Applying a chat template to a base model not fine-tuned for it degrades reasoning, yet appending RSPs recovers up to +29 pp on

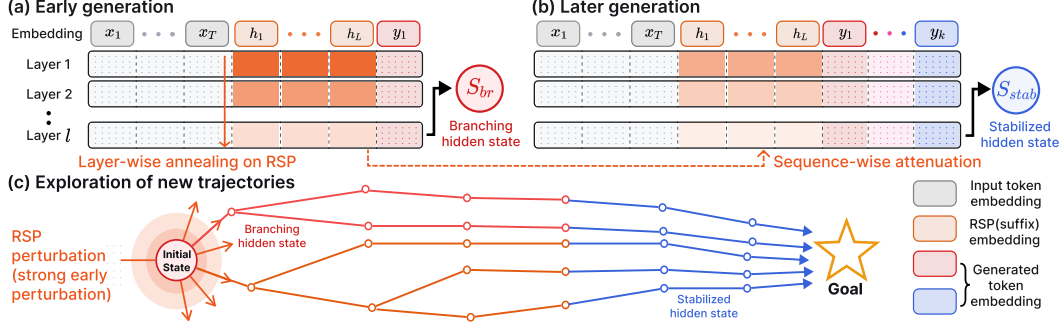


Figure 1: Conceptual overview of RSP-induced trajectory diversity. The hidden states shown are the final-layer outputs that drive the next-token distribution. (a) Early decoding: RSP perturbation reaches a *branching state* (red; formalized in §3.1), whose top- K differs from the no-RSP baseline and induces a diverse output distribution. (b) Later decoding: as the KV cache grows, the bound on RSP attention mass narrows (Theorem 1); branching events become rare and the hidden state (blue) commits to the branch already opened. (c) Across rollouts, the trajectories opened by early RSP perturbation accumulate into independent paths.

37 Qwen2.5-Math-1.5B; if simple noise were merely shaking the model, accuracy should have worsened
38 instead. The injection itself, independent of any learned content, alters model behavior.

39 We analyze training-free Random Soft Prompts (RSPs) on LLM reasoning. With each rollout
40 receiving an *independent* draw, a single RSP injection *branches* hidden states into diverse trajectories
41 early on, which then *stabilize* as decoding proceeds. Empirically, RSPs share the early-decoding
42 entropy signature of optimized methods (LTPO, TTSV, SoftCoT) despite carrying no learned content;
43 Pass@ N accumulates only under *independent* resampling; and the same input-side diversity composes
44 with DAPO [Yu et al., 2025] training. Our contributions are: (i) RSP isolates the simplest form of
45 soft prompt — training-free and freshly resampled per rollout — providing a unified lens for the
46 structural effect of injection across variants that otherwise differ in training objective and prompt form,
47 while still reaching accuracy comparable to optimized methods in several settings; (ii) a theoretical
48 and empirical account of the mechanism (attention-mediated branch opening, isotropic directional
49 coverage, KV-cache attenuation); and (iii) DAPO extension showing independent RSP resampling
50 composes with rollout-diversity training.

51 2 Background

52 2.1 Soft Prompt Methods for LLM Reasoning

53 Existing *soft prompt* methods all share a common structure: they inject out-of-vocabulary continuous
54 embeddings and *optimize* them under task-specific objectives. Because evaluation is anchored to
55 end-task accuracy, the contributions of “injection itself” and “optimization” are not separated, and
56 how plain embedding injection reshapes model behavior remains an open question. As parameter-
57 efficient alternatives to full fine-tuning, Prefix-Tuning [Li and Liang, 2021], Prompt Tuning [Lester
58 et al., 2021], and P-tuning [Liu et al., 2024] reached near-fine-tuning performance using learnable
59 continuous vectors, after which LLM-reasoning-specific variants quickly followed. TTSV [Kang
60 et al., 2026] optimizes prefix embeddings at test time via trajectory entropy minimization; SoftCoT
61 [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant
62 model; LTPO [Ye et al., 2026] optimizes the soft prompt per problem to maximize its own confidence
63 on that problem; COCONUT [Hao et al., 2025] feeds the model’s own hidden states back as input
64 embeddings; MemGen [Zhang et al., 2026] uses embedding vectors as experience memory; and
65 Pause tokens [Goyal et al., 2024] insert learnable tokens to provide additional computation steps.

66 2.2 Noise Injection in Neural Networks

67 Neural networks have incorporated noise at several stages and locations. During training, dropout
68 [Srivastava et al., 2014] randomly deactivates *hidden activations*, and NEFTune [Jain et al., 2024]

69 adds Gaussian noise to *input embeddings* during instruction fine-tuning to improve instruction
70 following. At inference time, randomized smoothing [Cohen et al., 2019] injects Gaussian noise into
71 *raw inputs* to obtain certified robustness, and SmoothLLM [Robey et al., 2025] perturbs prompts at
72 the *discrete character* level to defend against jailbreaking. In RL, NoisyNet [Fortunato et al., 2018]
73 adds learnable, trained noise to *network weights* to aid exploration.

74 RSP appends noise to *embeddings*, and differs from these in three respects. (1) It operates without
75 any training, purely at inference, and does not modify model parameters (unlike dropout, NEFTune,
76 and NoisyNet). (2) It preserves the original input and appends the embedding within a single forward
77 pass, so it does not require the “multiple noisy passes plus aggregation” that robustness methods
78 need. (3) It uses vectors drawn directly from the pretrained embedding statistics, with no learned
79 parameters at all (NoisyNet, by contrast, learns its noise parameters).

80 3 Empirical Validation

81 RSP injection produces three empirical patterns: it preserves and sometimes recovers baseline
82 accuracy (§3.3); it reshapes early-generation entropy as Figure 1 predicts (§3.4); and Pass@ N
83 accumulates only under *independent* resampling (§3.5). §5 explains these mechanistically.

84 3.1 Random Soft Prompt

85 To isolate the act of injection from learned content while keeping the magnitude comparable to
86 real tokens and not privileging any direction in embedding space, we draw RSP from an isotropic
87 Gaussian fitted to the entrywise statistics of the pretrained embedding table. Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote
88 the pretrained token-embedding matrix (V vocab size, d hidden dim). Write its entrywise statistics as
89 $\mu_E := \frac{1}{V \cdot d} \sum_{v,k} [\mathbf{W}_E]_{v,k}$ and $\sigma_E := \text{std}(\mathbf{W}_E)$. An RSP of length L is a sequence of continuous
90 vectors $\mathbf{H} = [h_1, \dots, h_L] \in \mathbb{R}^{L \times d}$ drawn independently from

$$h_j \sim \mathcal{N}(\mu_E \mathbf{1}_d, \sigma_E^2 \mathbf{I}_d), \quad j = 1, \dots, L. \quad (1)$$

91 The centered form $\bar{h}_j := h_j - \mu_E \mathbf{1}_d \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$ is a zero-mean isotropic Gaussian. With
92 $\mathbf{X} \in \mathbb{R}^{T \times d}$ the input token-embedding sequence (T prompt tokens, rows from $\mathbf{W}_E$), the main text
93 uses the *suffix* form $[\mathbf{X}; \mathbf{H}] \in \mathbb{R}^{(T+L) \times d}$; other positions (prefix, infix [Kang et al., 2026, Xu et al.,
94 2025, Ye et al., 2026, Zhang et al., 2026]) are reported as ablations in §3.3 and Appendix A. RSP is
95 not learnable, so it receives no gradient and is freshly drawn per rollout.

96 Call decoding step t a *branching event* and the last-layer hidden state $\mathbf{s}^{(t)}$ a *branching state* when its
97 top- K differs from the no-RSP baseline $\bar{\mathbf{s}}^{(t)}$ (Figure 1, red).¹ §5 uses this terminology.

98 3.2 Experimental Setup

99 We evaluate on three mathematical reasoning benchmarks, MATH-500 [Lightman et al., 2024],
100 GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024]
101 and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and
102 base models with mismatched chat templates. Tables 1 and 2 use greedy decoding to isolate the effect
103 of RSPs from sampling stochasticity; the diversity and Pass@ N experiments in §3.5 use the sampling
104 settings specified there. RSPs follow Eq. (1), using the default *suffix* position with a freshly drawn
105 $\mathbf{H}$ per rollout. Since Theorem 1 identifies the RSP length L as the design parameter capping the
106 maximum perturbation influence, we ablate $L \in \{10, 15, 20\}$ per model to identify an appropriate
107 perturbation strength. Tables 1 and 2 report results at the per-model selected length. The mechanism
108 analyses in §3.4–§3.5 use a fixed setting ($L=10$, *suffix*), which insulates them from this selection
109 effect. Full experimental details and per-position results are in Appendix A.

110 3.3 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

111 We first examine how injecting random embeddings affects reasoning performance. Table 1 reports
112 accuracy at the default *suffix* position alongside *prefix* as a position ablation.

¹Italic h_j : RSP input. Bold $\mathbf{s}^{(t)}$: hidden state. K denotes the cardinality in top- K analyses; nucleus sampling uses a separate threshold p . t indexes decoding step, ℓ self-attention layer.

Table 1: Accuracy (%) across model configurations and benchmarks. Values in parentheses indicate the difference from the baseline. Full per-position breakdown (including *infix*) is in Appendix A.

Model	MATH-500			GSM8K			AIME24		
	Baseline	RSP(<i>prefix</i>)	RSP(<i>suffix</i>)	Baseline	RSP(<i>prefix</i>)	RSP(<i>suffix</i>)	Baseline	RSP(<i>prefix</i>)	RSP(<i>suffix</i>)
<i>Instruct Models</i>									
LLaMA-3.1-8B-Inst	52.20	45.00 (−7.2)	49.40 (−2.8)	85.75	76.35 (−9.4)	86.73 (+1.0)	6.67	3.33 (−3.3)	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	81.40 (−1.8)	83.40 (+0.2)	95.45	94.92 (−0.5)	95.68 (+0.2)	13.33	13.33 (+0.0)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	74.80 (+1.8)	74.20 (+1.2)	85.29	85.29 (+0.0)	84.69 (−0.6)	10.00	13.33 (+3.3)	20.00 (+10.0)
<i>Base Models (Raw Text)</i>									
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	55.60 (−16.6)	84.15	84.31 (+0.2)	54.13 (−30.0)	6.67	16.67 (+10.0)	13.33 (+6.7)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	54.60 (−6.8)	77.86	74.30 (−3.6)	58.61 (−19.3)	16.67	10.00 (−6.7)	3.33 (−13.3)
<i>Base Models + ChatML (Format Mismatch)</i>									
Qwen2.5-Math-7B	52.20	59.20 (+7.0)	70.40 (+18.2)	58.30	56.41 (−1.9)	75.97 (+17.7)	23.33	3.33 (−20.0)	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	51.00 (+16.6)	37.45	67.02 (+29.6)	50.64 (+13.2)	13.33	20.00 (+6.7)	13.33 (+0.0)

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. **Baseline** from Table 1; **RSP** is the best over positions and $L \in \{10, 15, 20\}$ (breakdown in Appendix A). **Bold**: best; underline: second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	84.20	83.80	82.80	83.00
	GSM8K	<u>95.45</u>	95.68	95.30	95.00	94.84
	AIME24	13.33	<u>16.67</u>	23.30	<u>16.67</u>	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	<u>75.20</u>	<u>75.20</u>	76.00	72.40
	GSM8K	85.29	85.75	<u>85.70</u>	84.46	84.53
	AIME24	<u>10.00</u>	20.00	6.70	6.67	<u>10.00</u>

On instruct models RSP stays within a few percentage points of the baseline. The most striking pattern is format-mismatch recovery: when a ChatML template is applied to a base model not trained for that format, *Suffix* recovers +18.2/ + 17.7 pp on Qwen2.5-Math-7B (MATH-500/GSM8K) and *prefix* recovers up to +29.0/ + 29.6 pp on Qwen2.5-Math-1.5B — if simple noise were responsible, accuracy should have dropped further. In the base raw-text setting the picture inverts: *suffix* incurs sharp losses (−16.6/ − 30.0 pp on Qwen2.5-Math-7B) while *prefix* changes accuracy less, suggesting base models treat EOS-adjacent signals as a derailment trigger that *prefix* avoids.

RSP’s *direct accuracy gains* concentrate in misaligned-input recovery and stay within $\pm$ a few percentage points on well-aligned instruct models. The effects this paper centers on are the token-, trajectory-, and outcome-level diversity in §3.5, with accuracy recovery as a secondary outcome.

We further compare RSP with TTSV, SoftCoT, and LTPO — prior soft prompt methods that optimize embeddings under different objectives. Table 2 reports a unified evaluation under the same fallback-based answer-extraction pipeline (Appendix A.2). RSP does not consistently dominate the trained methods — TTSV, for instance, beats RSP on AIME24 with Qwen2.5-Math-7B-Instruct (23.30% vs 16.67%). What we want to highlight is precisely the converse claim: even without training, RSP reaches a comparable range, and that fact is itself this paper’s main evidence that part of the soft-prompt effect comes from the *structure of injection* rather than the learned content.

3.4 How Does RSP Affect the Output Distribution?

We next examine how RSP injection reshapes the output distribution. The setting is Qwen2.5-Math-1.5B-Instruct on MATH-500, and we measure per-token entropy, top-1 probability, and varentropy across generation steps. Figure 2 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV); full curves are in Appendix A.4.

Latent prompt methods including RSP share the same signature: *a rise in early-generation entropy followed by convergence to the baseline later in generation*. A natural mechanistic explanation is the out-of-vocabulary (OOV) effect — when continuous embeddings outside the vocabulary enter the input, the model has to attend to a never-before-seen key position on top of its familiar token distribution, redistributing probability mass across multiple candidates within the early next-token distribution. All three RSP injection positions show early-stage entropy $\uparrow$, top-1 probability $\downarrow$, and

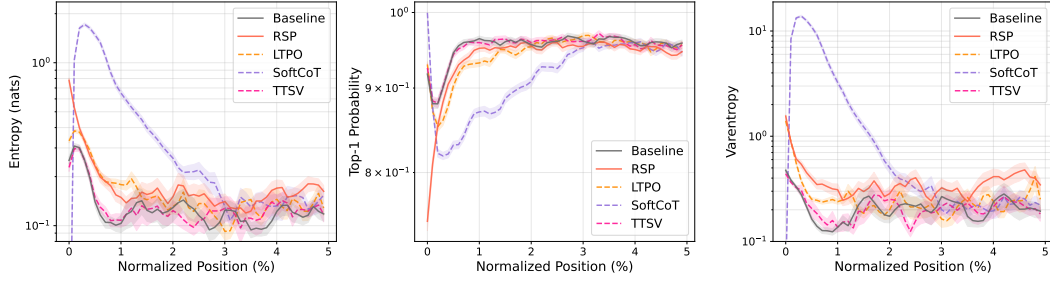


Figure 2: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

varentropy $\uparrow$, jointly consistent with multimodal distributions [Ahmed et al., 2026]. The change is confined to the early stage and all three metrics return to baseline in the middle and final portions, consistent with the *early influence with automatic KV cache-growth decay* pattern that §5.2 will derive theoretically.

Prior methods trained under different objectives (LTPO, SoftCoT, TTSV) share this signature. Even TTSV, whose reward explicitly *lowers* mean reasoning-trajectory entropy, shows early-stage entropy comparable to the baseline. The entropy value itself is not a quality score — higher entropy is not always better; rather, the signature reveals a *shared mechanism* in which OOV embedding injection produces the same fingerprint independently of the optimization objective, supporting the hypothesis that the key driver of latent reasoning is the *act of injection*, not the learned content.

3.5 Does RSP Induce Trajectory Diversity?

Section 3.4 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP’s first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, *suffix* injection, and $L = 10$.

Token-level: distribution beyond temperature.

We quantify how much RSP shifts the first-token distribution — the critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at $\tau=1.0$ as reference, we compare baselines at $\tau=2.0, 3.0$ (16 samples per problem) against RSP at $\tau=1.0$ (averaged over 16 seeds) on three metrics: Spearman rank correlation ρ to detect rank reorganization, the probability mass placed outside the baseline’s top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change (formal definitions in Appendix B). Table 3 shows the contrast: RSP’s distributional shift falls between $\tau=2.0$ and $\tau=3.0$ in magnitude, but the mechanism differs — temperature is a monotone transform that preserves ranking ($\rho = 1$ at any τ), while RSP partially preserves but reranks ($\rho = 0.709$). The Pass@1 row exposes the cost: matching RSP’s magnitude with temperature toward $\tau=3.0$ collapses accuracy to 1.42%, whereas RSP preserves 73.30%. Thus, RSP produces a fundamentally different perturbation than temperature scaling. Moreover, RSP’s accuracy variability across 16 seeds at $\tau=1.0$ is ± 1.07 , comparable to the ± 1.47 that temperature sampling produces across 16 samples at $\tau=2.0$.

Table 3: First-token distribution metrics measured against the baseline at $\tau=1.0$, comparing baselines at higher temperatures ($\tau=2.0, 3.0$) with RSP at $\tau=1.0$. RSP metrics are averaged over 16 random seeds; Acc reports mean $\pm$ std across 16 samples per problem.

Metric	$\tau=2.0$	$\tau=3.0$	RSP
Spearman ρ	1.000	1.000	0.709
Mass outside top-10	5.18%	29.11%	21.86%
JS divergence	0.060	0.218	0.131
Acc (%)	20.77 ± 1.47	1.42 ± 0.35	73.30 ± 1.07

Trajectory-level: semantic diversity.

Token-level reranking raises a second question: do first-token perturbations diverge

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	Baseline	RSP
Pairwise cos. dist.	0.0612	0.0879
Inter-cluster dist.	0.0183	0.0982
Intra-cluster dist.	0.0526	0.0528

180 into distinct trajectories, or converge to similar ones? We sample
181 64 problems from MATH-500 and generate 300 independent
182 trajectories per problem under Baseline and RSP at $\tau = 1.0$.
183 Each trajectory is encoded by BGE-M3 [Chen et al., 2024] into a
184 dense semantic vector, and we compute three metrics: *pairwise*
185 *cosine distance* (overall semantic spread), *inter-cluster distance*
186 between centroids of DBSCAN [Ester et al., 1996], a density-
187 based clustering algorithm (separation between distinct trajectory groups), and *intra-cluster distance*
188 (coherence within groups). Table 4 reveals three concurrent patterns: pairwise distance rises by
189 $1.4\times$, inter-cluster distance jumps by $5.4\times$, and intra-cluster distance is essentially unchanged —
190 the trajectory set becomes more diverse and splits into more separated groups while preserving
191 within-group coherence. RSP also yields larger pairwise distance than baseline on 56 of 64 problems.

192 **Outcome-level: Pass@ N scaling.** Does this token- and trajectory-level diversity translate into
193 task-level outcome diversity, measured by Pass@ N [Chen et al., 2021] (the probability that at least
194 one of N sampled solutions is correct)? We generate 16 samples per problem on MATH-500 and
195 AIME24 and compare four conditions: (i) *Baseline* at $\tau = 1.0$; (ii) *RSP (single seed)* at $\tau = 1.0$, with
196 one RSP shared across all samples; (iii) *RSP (indep. seed)* with greedy decoding, a fresh RSP per
197 sample; and (iv) *RSP (indep. seed)* at $\tau = 1.0$, a fresh RSP per sample.

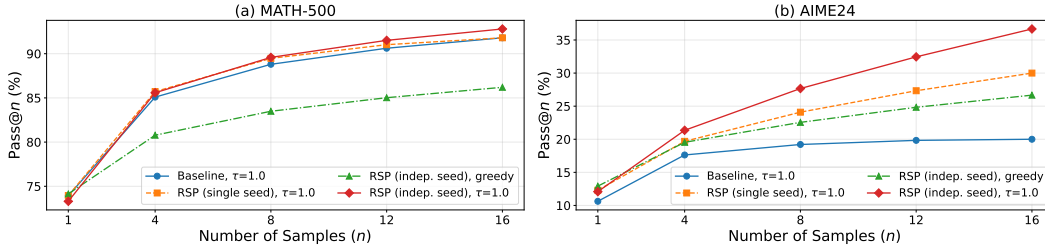


Figure 3: Pass@ N scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, 16 samples per problem. *Baseline*: temperature sampling only. *RSP (single seed)*: single RSP shared across samples combined with temperature. *RSP (indep. seed)*: a different RSP per sample, with or without temperature.

198 Condition (iv) achieves the highest Pass@ N on both benchmarks, reaching 92.80% on MATH-
199 500 and 36.67% on AIME24 at $N = 16$, while (ii) shows no improvement over the baseline —
200 the diversity gain depends on varying the embeddings across samples rather than fixing a single
201 perturbation. Pass@1 stays comparable to the baseline on MATH-500, and on the harder AIME24
202 (iv) even surpasses it, suggesting that independent RSPs combined with temperature sampling expand
203 trajectory diversity without sacrificing single-sample accuracy.

204 4 Application: DAPO Training with RSP

205 §3 showed at inference time that RSP yields Pass@ N gains tied to independent resampling and
206 rank-changing perturbations unattainable by temperature; the mechanism behind these patterns is
207 analyzed later in §5. Here we ask whether the same effect carries over to training time. Rollout
208 diversity drives learning in sampling-based RL, and DAPO [Yu et al., 2025] is a recent GRPO [Shao
209 et al., 2024] variant explicitly designed to preserve diversity at the loss level. We use it as a stringent
210 testbed: does input-side branch opening compose with loss-side preservation to yield further gains?
211 Details and per-step results are in Appendix F.

212 DAPO preserves rollout diversity at the loss level by reshaping how rollouts are weighted and
213 clipped during long chain-of-thought RL; RSP induces diversity *before* the loss is computed by
214 perturbing the rollout distribution at the input level (§3.5). Writing each rollout’s token sequence as
215 $y_i = (y_{i,1}, \dots, y_{i,|y_i|})$ (using y to avoid collision with the head output $\mathbf{o}_i^{(\ell)}$ of §5.1) and pairing each
216 rollout with an independently sampled RSP $\mathbf{H}_i$ (reused at update time to keep the update on-policy),
217 importance ratio $r_{i,t}(\theta) = \pi_\theta(y_{i,t} \mid q, \mathbf{H}_i, y_{i,<t}) / \pi_{\theta_{\text{old}}}(y_{i,t} \mid q, \mathbf{H}_i, y_{i,<t})$, group-relative advantage

218 $\hat{A}_{i,t}$, and asymmetric clip range $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$, the DAPO + RSP objective is

$$\mathcal{J}_{\text{DAPO+RSP}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_i |y_i|} \sum_{i,t} \min \left(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon_{\text{low}}, 1+\varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \right]. \quad (2)$$

Table 5: Per-benchmark accuracy (%) at each method’s peak step on Qwen2.5-Math-7B. **Bold** denotes the better of the two RL methods on each benchmark. Avg is the unweighted five-benchmark mean.

Method	GSM8K	MATH-500	College	Minerva	AIME 24	Avg
Base	57.2	52.6	18.3	13.6	8.6	30.06
DAPO	86.3	78.2	41.4	37.5	22.6	53.20
DAPO + RSP	87.7	78.6	42.2	39.7	23.6	54.36

219 We train Qwen2.5-Math-7B on a MATH Level-3–5 subset (~8.5K prompts) with SimpleRL-Zoo
 220 [Zeng et al., 2025] under Eq. (2): each step uses $G = 8$ rollouts per prompt at $\tau=1.0$, batch 1,024,
 221 4 PPO mini-batches of 256, for 100 steps. *DAPO* is compared against *DAPO + RSP*, where suffix
 222 RSP ($L=20$) is applied during rollouts only and evaluation uses no RSP. We evaluate on five math
 223 benchmarks (GSM8K [Cobbe et al., 2021], MATH-500 [Lightman et al., 2024], College Math,
 224 Minerva Math [Lewkowycz et al., 2022], AIME 2024) and report the unweighted mean; AIME 2024
 225 uses Avg@32 at $\tau=1.0$, the others greedy.

226 Figure 4 and Table 5 show two patterns: a late-
 227 stage advantage and a delayed early reward growth.
 228 DAPO + RSP reaches 54.36% at step 90 (vs DAPO’s
 229 53.20% peak, +1.16 pp) and widens the gap to +3.10 pp
 230 at step 100 with DAPO’s post-peak decline delayed; ear-
 231 lier the curves cross around step 20. The patterns follow
 232 from the methods acting at separate stages: DAPO pre-
 233 serves diversity at the loss level over already-generated
 234 rollouts while RSP perturbs the hidden states that pro-
 235 duce them (§5.1), giving the policy broader learning
 236 signals — input-side exploration with loss-side preserva-
 237 tion. These results use a single training seed and should
 238 be read as a feasibility demonstration; multi-seed eval-
 239 uation is future work.

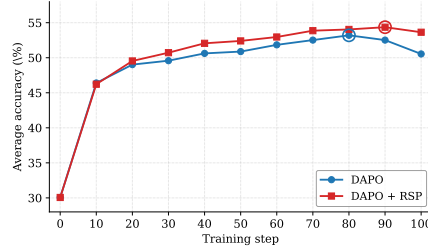


Figure 4: Five-benchmark average accuracy on Qwen2.5-Math-7B. DAPO + RSP reaches a higher peak (step 90) and stays stable through step 100.

240 5 Why RSP Works

241 Table 1’s headline — random embeddings without training already lift reasoning accuracy — raises
 242 one question: *why* does an untrained random vector help? §3–§4 added four mechanistic patterns:
 243 early entropy rise, trajectory-level diversification, Pass@ N accumulation under independent draws,
 244 and a late-training DAPO advantage. RSP enters the hidden state through self-attention, so within
 245 one head the magnitude of its instantaneous contribution reduces to one scalar: the *attention mass* w
 246 on RSP tokens. Early in decoding w is large enough to branch hidden states into multiple directions
 247 (§5.1); as the KV cache grows, the upper bound on w narrows automatically and the model commits
 248 to the branch already opened (§5.2) — an implicit explore-then-exploit. Independent resampling per
 249 rollout opens a fresh branch each time (§3.1 for RSP definition and *branching event/state* terminology;
 250 proofs in Appendix D).

251 5.1 Exploration: RSP strengthens early-stage exploration

252 Inserting a random prompt concentrates attention on it, and that concentration is what amplifies
 253 exploration. The effect of a single injection on the hidden state raises two questions — *how much*
 254 and *where* — decided respectively by the scalar w inside one attention head and by the distribution of
 255 $\mathbf{H}$. The isotropic Gaussian addresses both at once.

256 Consider one attention head in self-attention layer ℓ . At query position i , the query attends $n \geq 1$
 257 unmasked real tokens together with $L \geq 1$ RSP tokens, all attention logits finite. Write the attention

weights as $\alpha_{ij}^{(\ell)}$ and the value vectors on the real and RSP sides as $\mathbf{v}_j^{(\ell)}, \mathbf{v}_{r,j}^{(\ell)}$. Defining the total
attention mass on random tokens $w_{r,i}^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)}$, the real-token mass $1 - w_{r,i}^{(\ell)}$ is positive
and the head output $\mathbf{o}_i^{(\ell)}$ splits *exactly* into a renormalized real-token term $\tilde{\mathbf{o}}_i^{(\ell)}$ and an RSP-induced
contribution $\boldsymbol{\eta}_i^{(\ell)}$:

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)}, \quad \tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^n \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)}, \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r,j}^{(\ell)}. \quad (3)$$

Derivation of Eq. (3). Write the head output as $\sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r,j}^{(\ell)}$, then factor
 $1 - w_{r,i}^{(\ell)} = \sum_{j=1}^n \alpha_{ij}^{(\ell)} > 0$ out of the real-token sum. No approximation beyond the softmax-
normalization identity is used. $\square$

This single quantity $w_{r,i}^{(\ell)}$ controls both the attenuation ratio $1 - w_{r,i}^{(\ell)}$ on the real signal and the upper
bound on the random contribution $\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_j \|\mathbf{v}_{r,j}^{(\ell)}\|$. The *magnitude* of RSP-induced
variation thus reduces to one number in $[0, 1]$. Early in decoding the KV cache is short and this value
is non-negligible, so a single RSP draw can perturb next-token decisions.

The scalar w controls *how much*; *where* is decided by the distribution of $\mathbf{H}$. For the local logit
argument, isolate one RSP position and let $\bar{h} \in \mathbb{R}^d$ denote the i.i.d. marginal of any centered vector
 $\bar{h}_j$ in Eq. (1), with the other centered RSP positions fixed at zero. Let $z_a(\bar{h})$ be the under-RSP output
logit at vocab token a and write the vocab-logit gap $\Delta_{ab}(\bar{h}) := z_a(\bar{h}) - z_b(\bar{h})$ ($\bar{h} = 0$ is the centered
mean, not the no-RSP state; Appendix C). The transformer’s logit map is non-linear in $\bar{h}$, but a
first-order Taylor expansion around $\bar{h} = 0$ gives the local surrogate $\Delta_{ab}(\bar{h}) \approx \Delta_{ab}(0) + b_{ab}^\top \bar{h}$, with
 $b_{ab} := \nabla_{\bar{h}}(z_a - z_b)|_{\bar{h}=0} \in \mathbb{R}^d$ the gradient direction along which vocab tokens a, b swap rank (not
unit-norm in general). Because RSP is training-free, we cannot know in advance which b_{ab} opens
a useful branch, so the distribution must avoid systematically under-perturbing any direction. A
deterministic injection pins to one direction; vocabulary sampling inherits the embedding table’s
anisotropy. The remaining design — maximize the worst-case directional variance at fixed budget —
is uniquely solved by isotropy.

Proposition 1 (Maximin directional coverage). *Let $\mathcal{D}_\rho$ denote the family of zero-mean distributions
on $\mathbb{R}^d$ whose covariance satisfies $\text{tr}(\Sigma_D) \leq \rho^2$. For each $D \in \mathcal{D}_\rho$, $\min_{\|u\|=1} \text{Var}_{h \sim D}(u^\top h) =$
 $\lambda_{\min}(\Sigma_D)$, and $\sup_{D \in \mathcal{D}_\rho} \lambda_{\min}(\Sigma_D) = \rho^2/d$, attained iff $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.*

This is a *design criterion* for the prompt law, not a guarantee of correctness; correctness enters
through the task-side $p_{\min}(x)$ assumption of §5.2 and independent resampling.²

Gaussian RSP attains the equality with $\Sigma = \sigma_E^2 \mathbf{I}_d$ ($\rho^2 = d\sigma_E^2$) and adds *full support* on $\mathbb{R}^d$
(Appendix C). Isotropy excludes no direction; full support gives positive probability to the open
branching-event region (§3.1) whenever it is non-empty (Appendix D.3.3). Monotone temperature
preserves ranking and cannot reach this region, so RSP’s *rank-changing* branching follows from the
two properties together, splitting the hidden state into multiple branching states early in reasoning.

5.2 Annealing: KV cache growth dilutes RSP attention

The early branching stabilizes as decoding proceeds: the upper bound on $w_{r,i}^{(\ell)}$ narrows along the
sequence axis with autoregressive cache growth. Each step adds one real-token term while RSP terms
stay fixed; at frozen attention-logit gap, this asymmetry yields the following bound.

Theorem 1 (Attention-mass decay under KV cache growth). *Consider a single attention head with
per-head key dimension d_{head} , where query $\mathbf{q}_i$ at position i attends $n \geq 1$ real keys $\mathbf{k}_j$ and $L \geq 1$
RSP keys $\mathbf{k}_{r,j}$, under finite logits. Define the pre-scale attention-logit gap $\Delta_i := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j -$*

²The maximin is stated in input space; isotropy is not claimed to survive transformer layers. Any logit direction pulls back
via the model Jacobian to an input-space b , and isotropy guarantees only that no such pulled-back direction (in the role of u
above) has zero projected variance.

298 $\max_{j' \in [L]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ (distinct from the vocab-logit gap Δ_{ab} of §5.1). In scaled dot-product attention,

$$w_{r,i} \leq \frac{L}{L + n \exp(\Delta_i / \sqrt{d_{\text{head}}})},$$

299 and the right-hand side is strictly decreasing in n and tends to 0 at fixed L and Δ_i .

300 *Proof.* Let $s_j := \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_{\text{head}}}$, $s_{r_{j'}} := \mathbf{q}_i^\top \mathbf{k}_{r_{j'}} / \sqrt{d_{\text{head}}}$, $s_{r,\max} := \max_{j'} s_{r_{j'}}$, and $R :=$
 301 $\sum_{j'=1}^L \exp(s_{r_{j'}})$. The gap definition gives $s_j \geq \Delta_i / \sqrt{d_{\text{head}}} + s_{r,\max}$ for every real j , and
 302 $\exp(s_{r,\max}) \geq R/L$ holds because the max dominates the average; combining the two and sum-
 303 ming over the n real keys yields $\sum_{j=1}^n \exp(s_j) \geq (n/L) \exp(\Delta_i / \sqrt{d_{\text{head}}}) R$. Substituting into
 304 $w_{r,i} = R / (\sum_j \exp(s_j) + R)$ gives

$$w_{r,i} \leq \frac{R}{(n/L) \exp(\Delta_i / \sqrt{d_{\text{head}}}) R + R} = \frac{L}{L + n \exp(\Delta_i / \sqrt{d_{\text{head}}})}.$$

305 The derivative of the right-hand side in n is $-L \exp(\Delta_i / \sqrt{d_{\text{head}}}) / (L + n \exp(\Delta_i / \sqrt{d_{\text{head}}}))^2 < 0$, so
 306 the bound is strictly decreasing and tends to 0 as $n \rightarrow \infty$. $\square$

307 Applied at each layer ℓ to $w_{r,i}^{(\ell)}$ with gap $\Delta_i^{(\ell)}$, L caps the maximum influence while n grows each
 308 step. The theorem guarantees only sequence-axis attenuation at fixed (layer, query, gap); the gap may
 309 sharpen at deeper layers, as suggested empirically by Figure 5, but that effect is outside the theorem’s
 310 scope. The accurate reading is “the attainable upper bound at a fixed gap decreases monotonically.”
 311 Because Eq. (3) ties the random contribution to the same scalar, branching-event reachability inherits
 312 this envelope — as the cache grows, events become rare and the response commits to the branch
 313 already opened, an implicit explore-then-exploit. Figure 5 visualizes this pattern along both axes.

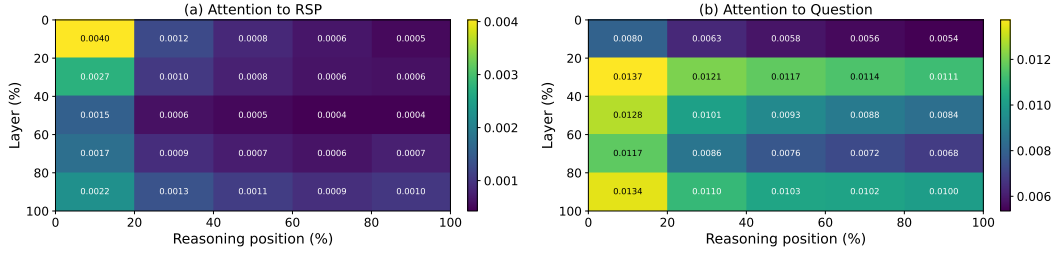


Figure 5: Per-token attention mass on Qwen2.5-Math-7B (*suffix*, 500 MATH-500 problems, independent RSP per problem). (a) RSP-token attention decreases along the reasoning-position axis, matching Theorem 1’s bound; the additional layer-axis decrease is empirical and outside the theorem’s scope. (b) Question-token attention stays comparatively uniform. Reported quantity: per-token mass (Appendix E), $w_{r,i}^{(\ell)} / L$ averaged over heads and samples.

314 Lifting single-shot branching to task accuracy across N rollouts requires (M1) marginal $p_{\min}(x) > 0$
 315 for a correct trajectory and (M2) mutual independence. Under both, $\Pr(\text{Pass}@N \text{ on } x) \geq 1 - (1 -$
 316 $p_{\min}(x))^N$ (Appendix D.3.4). §5.1 supports the branch-opening mechanism behind (M1) without
 317 guaranteeing correctness; *independent resampling* supplies (M2). Sharing one RSP breaks (M2) and
 318 removes the accumulation — the signature §3.5 documents empirically.

319 6 Conclusion

320 We studied Random Soft Prompts (RSPs), training-free isotropic Gaussian vectors fitted to pretrained
 321 embedding statistics. Latent-prompt methods share the same entropy signature, the gain vanishes
 322 when one RSP is shared across samples (§3), DAPO + RSP yields a modest late-training advantage
 323 (§4), and the per-layer RSP contribution is controlled by an attention-mass scalar whose fixed-gap
 324 upper bound shrinks under KV cache growth (§5). This signature emerges from purely random
 325 input-side perturbation, suggesting part of what trained soft prompts deliver is a *structural by-product*
 326 of injection and that RSP composes orthogonally with sampling-based pipelines as a near-zero-cost
 327 diversity expander.

References

- Farhan Ahmed, Yuya Jeremy Ong, and Chad DeLuca. LogitScope: A Framework for Analyzing LLM Uncertainty through Information Metrics. In *ICLR Workshop*, 2026.
- Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting Latent Predictions from Transformers with the Tuned Lens. *arXiv:2303.08112*, 2023.
- Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. In *Findings of ACL*, 2024.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating Large Language Models Trained on Code. *arXiv:2107.03374*, 2021.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training Verifiers to Solve Math Word Problems. *arXiv:2110.14168*, 2021.
- Jeremy M. Cohen, Elan Rosenfeld, and J. Zico Kolter. Certified Adversarial Robustness via Randomized Smoothing. In *ICML*, 2019.
- Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise. In *KDD*, 1996.
- Meire Fortunato, Mohammad Gheshlaghi Azar, Bilal Piot, Jacob Menick, Ian Osband, Alex Graves, Vlad Mnih, Remi Munos, Demis Hassabis, Olivier Pietquin, Charles Blundell, and Shane Legg. Noisy Networks for Exploration. In *ICLR*, 2018.
- Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer Feed-Forward Layers Are Key-Value Memories. In *EMNLP*, 2021.
- Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh Nagarajan. Think before You Speak: Training Language Models with Pause Tokens. In *ICLR*, 2024.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelfer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua

379 Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani,
 380 Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz
 381 Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yearly, Laurens van der
 382 Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo,
 383 Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat
 384 Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya
 385 Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman
 386 Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang,
 387 Olivier Duchenne, Onur Celebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic,
 388 Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu,
 389 Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira
 390 Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain
 391 Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar
 392 Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov,
 393 Shao-liang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale,
 394 Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane
 395 Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha,
 396 Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal
 397 Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gouget,
 398 Virginie Do, Vish Vogeti, Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin
 399 Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide
 400 Xia, Xinfeng Xie, Xuchao Jia, Xuwei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei,
 401 Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan,
 402 Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey,
 403 Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma,
 404 Alex Boesenberg, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo,
 405 Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew
 406 Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita
 407 Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh
 408 Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola,
 409 Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence,
 410 Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu,
 411 Changan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris
 412 Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel
 413 Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich,
 414 Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine
 415 Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban
 416 Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippos Kokkinos, Firat
 417 Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella
 418 Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi Zhang,
 419 Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha,
 420 Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan
 421 Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai
 422 Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya,
 423 Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica
 424 Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan
 425 Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal,
 426 Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran
 427 Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A,
 428 Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca
 429 Wehrstedt, Madian Khabza, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson,
 430 Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally,
 431 Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov,
 432 Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat,
 433 Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White,
 434 Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich
 435 Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem
 436 Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager,
 437 Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang,

438 Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra,
 439 Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ
 440 Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh,
 441 Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji
 442 Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Sheng Feng, Shenghao Lin,
 443 Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang,
 444 Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe,
 445 Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny
 446 Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara
 447 Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou,
 448 Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish
 449 Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov,
 450 Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian
 451 Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi,
 452 Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu Wang, Yu Zhao, Yuchen Hao,
 453 Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu
 454 Yang, Zhiwei Zhao, and Zhiyu Ma. The Llama 3 Herd of Models. *arXiv:2407.21783*, 2024.

455 Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian.
 456 Training Large Language Models to Reason in a Continuous Latent Space. In *ICLR*, 2025.

457 Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Thai, Junhao Shen, Jinyi Hu, Xu Han,
 458 Yujie Huang, Yuxiang Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. OlympiadBench: A
 459 Challenging Benchmark for Promoting AGI with Olympiad-Level Bilingual Multimodal Scientific
 460 Problems. In *ACL*, 2024.

461 Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea
 462 Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-Efficient Transfer Learning for NLP.
 463 In *ICML*, 2019.

464 Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang,
 465 and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models. In *ICLR*, 2022.

466 Neel Jain, Ping-yeh Chiang, Yuxin Wen, John Kirchenbauer, Hong-Min Chu, Gowthami Somepalli,
 467 Brian R. Bartoldson, Bhavya Kailkhura, Avi Schwarzschild, Aniruddha Saha, Micah Goldblum,
 468 Jonas Geiping, and Tom Goldstein. NEFTune: Noisy Embeddings Improve Instruction Finetuning.
 469 In *ICLR*, 2024.

470 Xinyue Kang, Diwei Shi, and Li Chen. Model Whisper: Steering Vectors Unlock Large Language
 471 Models’ Potential in Test-time. In *AAAI*, 2026.

472 Brian Lester, Rami Al-Rfou, and Noah Constant. The Power of Scale for Parameter-Efficient Prompt
 473 Tuning. In *EMNLP*, 2021.

474 Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay
 475 Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, Yuhuai Wu, Behnam
 476 Neyshabur, Guy Gur-Ari, and Vedant Misra. Solving Quantitative Reasoning Problems with
 477 Language Models. In *NeurIPS*, 2022.

478 Xiang Lisa Li and Percy Liang. Prefix-Tuning: Optimizing Continuous Prompts for Generation. In
 479 *ACL-IJCNLP*, 2021.

480 Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike,
 481 John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s Verify Step by Step. In *ICLR*, 2024.

482 Xiao Liu, Yanan Zheng, Zhengxiao Du, Ming Ding, Yujie Qian, Zhilin Yang, and Jie Tang. GPT
 483 Understands, Too. *AI Open*, 5:208–215, 2024.

484 Aman Madaan and Amir Yazdanbakhsh. Text and Patterns: For Effective Chain of Thought, It Takes
 485 Two to Tango. *arXiv:2209.07686*, 2022.

486 Alexander Robey, Eric Wong, Hamed Hassani, and George J. Pappas. SmoothLLM: Defending Large
 487 Language Models Against Jailbreaking Attacks. *Transactions on Machine Learning Research*,
 488 2025.

489 Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang,
490 Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the Limits of
491 Mathematical Reasoning in Open Language Models. *arXiv:2402.03300*, 2024.

492 Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng,
493 Haibin Lin, and Chuan Wu. HybridFlow: A Flexible and Efficient RLHF Framework. In *EuroSys*,
494 2025.

495 Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov.
496 Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine*
497 *Learning Research*, 15:1929–1958, 2014.

498 Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2
499 edition, 2018.

500 Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen,
501 Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen
502 Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 Rule: High-Entropy Minority Tokens Drive
503 Effective Reinforcement Learning for LLM Reasoning. In *NeurIPS*, 2025.

504 Yige Xu, Xu Guo, Zhiwei Zeng, and Chunyan Miao. SoftCoT: Soft Chain-of-Thought for Efficient
505 Reasoning with LLMs. In *ACL*, 2025.

506 An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu,
507 Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu,
508 Xingzhang Ren, and Zhenru Zhang. Qwen2.5-Math Technical Report: Toward Mathematical
509 Expert Model via Self-Improvement. *arXiv:2409.12122*, 2024.

510 Wengao Ye, Yan Liang, and Lianlei Shan. Thinking on the Fly: Test-Time Reasoning Enhancement
511 via Latent Thought Policy Optimization. In *ICLR*, 2026.

512 Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian
513 Fan, Gaohong Liu, Juncai Liu, Lingjun Liu, Xin Liu, Haibin Lin, Zhiqi Lin, Bole Ma, Guangming
514 Sheng, Yuxuan Tong, Chi Zhang, Mofan Zhang, Ru Zhang, Wang Zhang, Hang Zhu, Jinhua Zhu,
515 Jiaze Chen, Jiangjie Chen, Chengyi Wang, Hongli Yu, Yuxuan Song, Xiangpeng Wei, Hao Zhou,
516 Jingjing Liu, Wei-Ying Ma, Ya-Qin Zhang, Lin Yan, Yonghui Wu, and Mingxuan Wang. DAPO:
517 An Open-Source LLM Reinforcement Learning System at Scale. In *NeurIPS*, 2025.

518 Weihao Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Keqing He, Zejun Ma, and Junxian He. SimpleRL-
519 Zoo: Investigating and Taming Zero Reinforcement Learning for Open Base Models in the Wild.
520 In *COLM*, 2025.

521 Guibin Zhang, Muxin Fu, and Shuicheng Yan. MemGen: Weaving Generative Latent Memory for
522 Self-Evolving Agents. In *ICLR*, 2026.

A Experimental Setup and Full Accuracy Breakdown

All experiments are conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-Math-1.5B variants). The RSP token count L is selected per (model, benchmark) pair from $\{10, 15, 20\}$ (Table 7); the per-position breakdown is in Table 6.

Table 6: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
<i>Instruct Models</i>				
LLaMA-3.1-8B-Instruct	Baseline	52.20	85.75	<u>6.67</u>
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Suffix	<u>49.40</u> (−2.8)	86.73 (+1.0)	10.00 (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	<u>13.33</u>
	Prefix	81.40 (−1.8)	94.92 (−0.5)	<u>13.33</u> (+0.0)
	Suffix	83.40 (+0.2)	95.68 (+0.2)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	<u>85.29</u>	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	<u>13.33</u> (+3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	20.00 (+10.0)
<i>Base Models (Raw Text)</i>				
Qwen2.5-Math-7B	Baseline	72.20	84.15	6.67
	Prefix	<u>71.60</u> (−0.6)	84.31 (+0.2)	16.67 (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	<u>13.33</u> (+6.7)
Qwen2.5-Math-1.5B	Baseline	<u>61.40</u>	77.86	16.67
	Prefix	64.40 (+3.0)	<u>74.30</u> (−3.6)	<u>10.00</u> (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
<i>Base Models + ChatML (Format Mismatch)</i>				
Qwen2.5-Math-7B	Baseline	52.20	58.30	23.33
	Prefix	59.20 (+7.0)	56.41 (−1.9)	<u>3.33</u> (−20.0)
	Suffix	70.40 (+18.2)	75.97 (+17.7)	23.33 (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	<u>13.33</u>
	Prefix	63.40 (+29.0)	67.02 (+29.6)	20.00 (+6.7)
	Suffix	<u>51.00</u> (+16.6)	<u>50.64</u> (+13.2)	<u>13.33</u> (+0.0)

Table 7: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

A.1 RSP Injection Positions and Prompt Templates

Below we show the prompt structure for each injection position across all model types. **Blue text** indicates RSP embeddings, and **purple text** indicates special tokens.

A.1.1 Prefix

RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

ChatML (Qwen Instruct / Base + ChatML).


```
[Random Embeddings ( $L$  tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
[Random Embeddings ( $L$  tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
[Random Embeddings ( $L$  tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
```

A.1.2 Infix

A description text and special tokens are inserted into the prompt. The special token positions are then replaced with RSP embeddings at the embedding level.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}

There are { $L$ } special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the { $L$ } special tokens: <special_token_1>...<special_token_ $L$ >
<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}

There are { $L$ } special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the { $L$ } special tokens:
<reserved_special_token_0>...
<reserved_special_token_ $L$ >
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

547 **Raw Text (Qwen Base).**

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the {L} special tokens: <special_token_1>...<special_token_L>

Please reason step by step, and put your final answer within \boxed{}
```

549 **A.1.3 Suffix**

550 For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token.
551 For raw text models, RSP embeddings are concatenated at the end of the prompt.

552 **ChatML (Qwen Instruct / Base + ChatML).**

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}[Random Embeddings (L tokens)]<|im_end|>
<|im_start|>assistant
```

554 **LLaMA Chat Template.**

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}[Random Embeddings (L tokens)]<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

556 **Raw Text (Qwen Base).**

```
{question}
Please reason step by step, and put your final answer within \boxed{}. [Random
Embeddings (L tokens)]
```

558 **A.2 Answer Extraction and Grading**

559 The final answer is extracted from the model output through a fallback chain. Each step is attempted
560 in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. "final answer is \$...\$. I hope" pattern (Minerva format)
2. \boxed{...}: last non-empty box is used; empty \boxed{} are skipped
3. Text following "the answer is"
4. Text following "final answer is"
5. Last number in the output (regex fallback)

562 Extracted answers are normalized by removing \$, \left, \right, and unit strings. Grading
563 is performed via symbolic equivalence checking: exact string match, numerical comparison
564 (`math.isclose, rel_tol=1e-4`), and SymPy symbolic simplification.

565 **A.3 Reproduced Prior Work Hyperparameters**

566 All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct,
567 evaluated with greedy decoding (temperature = 0) and our unified answer extraction pipeline.

TTSV. *Prefix* length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 8 reports the per-benchmark hyperparameters.

Table 8: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 6 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 9 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 9: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle/last-10% segment means with weights 0.1/0.8/0.1. Extension of Figure 2 (main text) and Figure 6.

Metric	Baseline	<i>Prefix</i>	<i>Infix</i>	<i>Suffix</i>	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

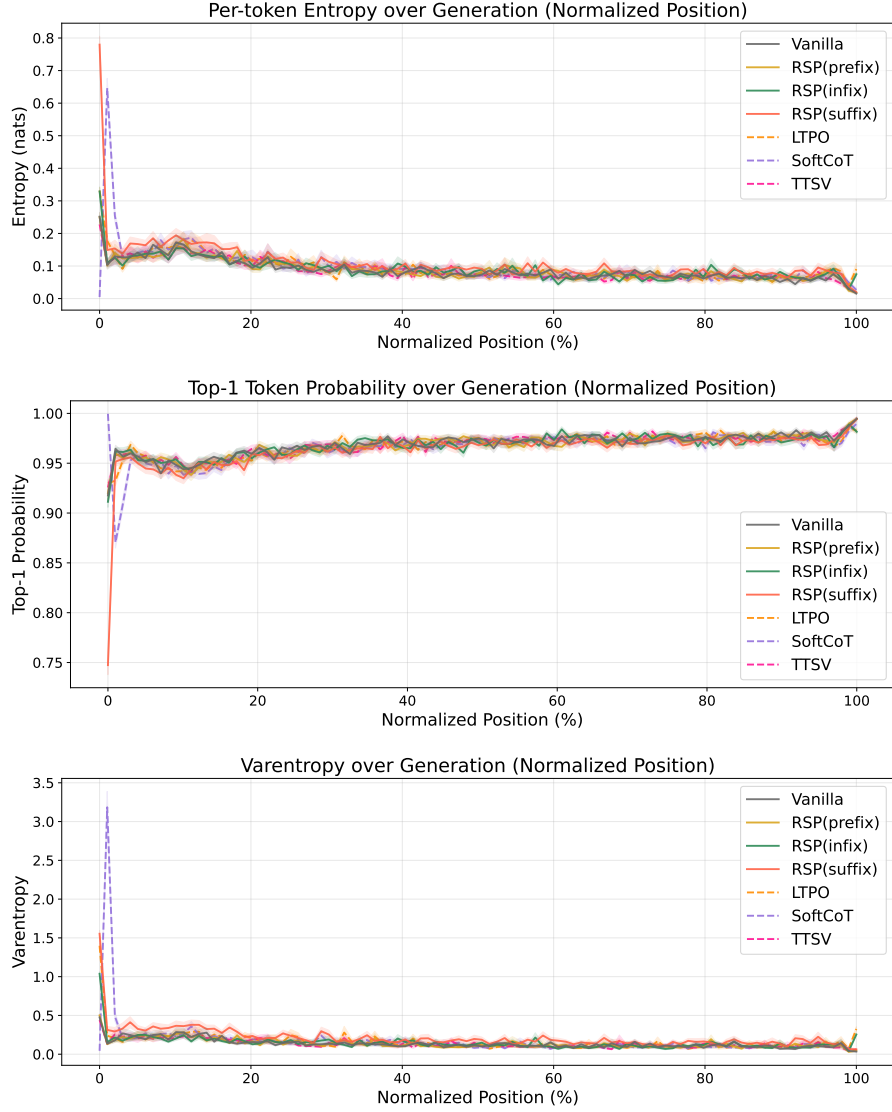


Figure 6: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior *soft prompt* methods. All methods converge after the initial stage.

584 B Token-Level Distribution Metrics

585 This appendix provides formal definitions for the metrics used in Section 3.5. Numerical results
 586 are reported in Table 3 of the main text. We extract first-token logits via a single forward pass (no
 587 autoregressive generation) and store the top-100 token ids and logit values per problem. Temperature
 588 conditions reuse the baseline logits scaled by $1/\tau$, requiring no separate inference. RSP draws
 589 16 independent seeds per problem (MATH-500), and reported metrics are the mean across the 16
 590 seeds. Pass@1 accuracy is computed over 16 samples per problem, with the baseline at $\tau=1.0$
 591 reaching $73.92 \pm 0.78\%$. Let P_{base} denote the baseline next-token distribution at $\tau=1.0$ and P_{target} the
 592 candidate distribution being compared (e.g., baselines at $\tau=2.0, 3.0$ or RSP at $\tau=1.0$). All metrics
 593 are computed analytically from saved logits at the first generation step.

594 **Spearman rank correlation.** Spearman’s ρ is the Pearson correlation coefficient between the token
 595 ranks of two distributions:

$$\rho = \text{Pearson}(\text{rank}(P_{\text{target}}), \text{rank}(P_{\text{base}})). \quad (4)$$

596 Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens
 597 is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore
 598 indicates rank reorganization beyond what temperature scaling can produce.

599 **Mass outside top- K .** Let $\text{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under
 600 P_{base} . The probability mass that the candidate distribution places *outside* this set is

$$\text{MassOutside}_K = 1 - \sum_{t \in \text{TopK}(P_{\text{base}})} P_{\text{target}}(t). \quad (5)$$

601 This directly measures support expansion: how much probability the candidate assigns to tokens that
 602 are not in the baseline’s preferred set. Reported values use $K = 10$.

603 **Jensen–Shannon divergence.** The symmetrized, bounded version of Kullback–Leibler divergence:

$$\text{JS}(P, Q) = \frac{1}{2} \text{KL}(P \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M), \quad M = \frac{1}{2}(P + Q). \quad (6)$$

604 We use base-2 logarithms so that $\text{JS} \in [0, 1]$. Tokens absent from the stored top-100 are assigned a
 605 sentinel logit (-10^9) before softmax, which is negligible for our setting.

C Prompt-Law Properties Used by the Theory

This appendix isolates the only properties of the RSP law used in the main text: centeredness, direction-agnostic covariance under a variance budget, and positive probability on every nonempty open set. We avoid stronger semantic claims because Section 5.1 needs only these geometric and measure-theoretic facts.

C.1 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let z_a^{Base} denote the no-RSP output logit at vocab token a and $z_a^{\text{RSP}}(\bar{h})$ the corresponding logit when the centered RSP perturbation takes value $\bar{h}$. The (true) vocab-logit gap under RSP is $\Delta_{ab}(\bar{h}) := z_a^{\text{RSP}}(\bar{h}) - z_b^{\text{RSP}}(\bar{h})$, distinct from the no-RSP baseline gap $\Delta_{ab}^{\text{Base}} := z_a^{\text{Base}} - z_b^{\text{Base}}$. Note that $\bar{h} = 0$ corresponds to the centered RSP at its entrywise mean rather than to the no-RSP forward pass, so in general $\Delta_{ab}(0) \neq \Delta_{ab}^{\text{Base}}$. The transformer’s logit map is non-linear in $\bar{h}$, but a first-order Taylor expansion around $\bar{h} = 0$ yields the local surrogate

$$\Delta_{ab}^{\text{lin}}(\bar{h}) := \Delta_{ab} + b_{ab}^\top \bar{h},$$

with $\Delta_{ab} := \Delta_{ab}(0)$ and $b_{ab} := \nabla_{\bar{h}}(z_a^{\text{RSP}} - z_b^{\text{RSP}})|_{\bar{h}=0}$. The proposition below is stated for the surrogate Δ_{ab}^{lin} . For the true non-linear gap $\Delta_{ab}(\bar{h})$, the first-order term has zero mean under centered perturbations, but the Taylor remainder $r_{ab}(\bar{h}) = O(\|\bar{h}\|^2)$ may contribute a second-order mean shift; we make no claim about its sign or magnitude.

Proposition 2 (No systematic first-order drift in the surrogate). *If $\mathbb{E}[\bar{h}] = 0$, then*

$$\mathbb{E}[\Delta_{ab}^{\text{lin}}(\bar{h})] = \Delta_{ab}$$

for every token pair (a, b) . If a deterministic centered prompt $\bar{h}_0$ is reused across runs, then

$$\Delta_{ab}^{\text{lin}}(\bar{h}_0) - \Delta_{ab} = b_{ab}^\top \bar{h}_0$$

is a fixed directional bias.

Proof. The centered case is immediate from linearity of expectation:

$$\mathbb{E}[\Delta_{ab}^{\text{lin}}(\bar{h})] = \Delta_{ab} + b_{ab}^\top \mathbb{E}[\bar{h}] = \Delta_{ab}.$$

The deterministic case follows by substitution. $\square$

This proposition rules out a run-averaged first-order bias. It does not say that individual prompt draws leave the logits unchanged.

C.2 Proof of Proposition 1

For a unit vector u , the first-order perturbation energy available along u is $\text{Var}_{h \sim D}(u^\top h) = u^\top \Sigma_D u$. The worst-case directional energy is therefore the minimum Rayleigh quotient of Σ_D .

Proof. For any symmetric covariance matrix Σ_D ,

$$\min_{\|u\|=1} u^\top \Sigma_D u = \lambda_{\min}(\Sigma_D).$$

Let the eigenvalues of Σ_D be $\lambda_1, \dots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \leq \frac{1}{d} \sum_{m=1}^d \lambda_m = \frac{\text{tr}(\Sigma_D)}{d} \leq \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. $\square$

The proof uses only covariance. We choose a Gaussian law for RSP because it adds the open-set reachability property proved next.

637 **C.3 Open-Set Reachability of Isotropic Gaussian Prompts**

638 **Proposition 3** (Open-set reachability). *Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ with $\sigma > 0$. Then every nonempty open*
 639 *set $U \subseteq \mathbb{R}^d$ satisfies*

$$\Pr(\bar{h} \in U) > 0.$$

640 *Proof.* The Gaussian density

$$(2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|\bar{h}\|^2}{2\sigma^2}\right)$$

641 is strictly positive at every point of $\mathbb{R}^d$. Every nonempty open set contains a ball of positive Lebesgue
 642 measure, and integrating a strictly positive density over that ball gives positive probability. $\square$

643 This is the only support property used in the informal branching argument of §5.1 and in Ap-
 644 pendix D.3.3.

645 D Proofs for the Transformer-Level Mechanism

646 This appendix follows the same order as the main theory section: exploration, annealing, and
647 performance gain. The prompt-law facts used before these proofs are collected in Appendix C.

648 D.1 Exploration: attention decomposition and perturbation size

649 **Proposition 4** (Attention decomposition). *For one attention head at query position i in layer ℓ , let*
650 *$\alpha_{ij}^{(\ell)}$ be the attention weights over $n \geq 1$ unmasked real tokens and $L \geq 1$ random tokens, with finite*
651 *attention logits. Define*

$$w_{r,i}^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)}$$

652 and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(\ell)} := \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}}.$$

653 Then

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)},$$

654 where

$$\tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^n \tilde{\alpha}_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} \quad \text{and} \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

655 *Proof.* The head output is

$$\mathbf{o}_i^{(\ell)} = \sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

656 By the softmax positivity and the existence of at least one unmasked real token, $\sum_{j=1}^n \alpha_{ij}^{(\ell)} =$
657 $1 - w_{r,i}^{(\ell)} > 0$, so

$$\sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \sum_{j=1}^n \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)}.$$

658 Substituting this identity gives the decomposition. □

659 **Corollary 1** (Magnitude scales with random-token attention). *For every realization of the random*
660 *values,*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

661 *Proof.* By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(\ell)}\| = \left\| \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)} \right\| \leq \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \|\mathbf{v}_{r_j}^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

662 □

663 Corollary 1 is the only quantitative fact needed in the main text. No independence assumption
664 between attention weights and random keys is required. Once $w_{r,i}^{(\ell)}$ is small, the random contribution
665 must be small as well.

666 D.2 Corollaries of Theorem 1 on KV cache growth

667 The fixed-gap decay bound yields the corollaries below.

668 **Corollary 2** (Sequence-wise annealing of the fixed-gap envelope). *For fixed L and $\Delta_i^{(\ell)}$, the upper*
 669 *bound*

$$f(n) = \frac{L}{L + n \exp(\Delta_i^{(\ell)} / \sqrt{d_{\text{head}}})}$$

670 *is strictly decreasing in n and tends to 0 as $n \rightarrow \infty$. KV cache growth alone therefore narrows the*
 671 *largest RSP attention mass compatible with that gap.*

672 During autoregressive decoding $\Delta_i^{(\ell)}$ also moves because queries, keys, and hidden states co-evolve.
 673 The accurate reading is not “the realized mass decreases at every step” but “the envelope compatible
 674 with a fixed gap shrinks as the KV cache grows.”

675 *Proof.* Differentiate $f(n)$ with respect to n :

$$f'(n) = -\frac{L \exp(\Delta_i^{(\ell)} / \sqrt{d_{\text{head}}})}{\left(L + n \exp(\Delta_i^{(\ell)} / \sqrt{d_{\text{head}}})\right)^2} < 0.$$

676 Thus $f(n)$ is strictly decreasing, and its denominator diverges linearly in n while its numerator is
 677 fixed, so $f(n) \rightarrow 0$. $\square$

678 **Corollary 3** (Vanishing random contribution under annealing). *If $n \rightarrow \infty$ while L and $\Delta_i^{(\ell)}$ remain*
 679 *fixed, $w_{r,i}^{(\ell)} \rightarrow 0$, and for every realization of the random values*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\| \rightarrow 0.$$

680 *Proof.* $w_{r,i}^{(\ell)} \rightarrow 0$ by Corollary 2. Apply Corollary 1 to bound $\|\boldsymbol{\eta}_i^{(\ell)}\|$. The maximum norm is finite
 681 for any realization of finitely many sampled values. $\square$

682 D.3 Performance gain: from local admission to Pass@N

683 From here we work at the vocab-logit and task levels, dropping the per-layer index ℓ . The main text
 684 uses a first-order surrogate to state two claims: the §5.1 claim that a baseline-excluded vocab token
 685 can enter the local top- K set, and the §5.2 closing that independent reruns turn such local openings
 686 into Pass@ N gains. The proofs below use only one support condition on the centered prompt law D :

$$(S1) \quad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

687 Proposition 3 shows that the isotropic Gaussian law of §5.1 satisfies (S1).

688 D.3.1 Autoregressive Formulation

689 In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^T p(y_t \mid x, y_{<t}),$$

690 where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by
 691 RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single
 692 global distribution.

693 D.3.2 Local Linearization of Logits under RSP

694 Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector from the RSP definition
 695 (Eq. (1) in §3.1), treated as a per-token surrogate. The actual implementation uses a length- L prompt
 696 $\mathbf{H} = (h_1, \dots, h_L)$ with L independent draws; the arguments below isolate how one local perturbation
 697 can shift the logits. We model the perturbed logits as a function $z_a(\bar{h})$ at vocab token a , assuming
 698 local smoothness in a neighborhood of $\bar{h} = 0$:

$$z_a(\bar{h}) = z_a(0) + \nabla_{\bar{h}} z_a(0)^\top \bar{h} + r_a(\bar{h}),$$

699 where $r_a(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $c_a := \nabla_{\bar{h}} z_a(0)$ and $z_a := z_a(0)$, we obtain the local affine
 700 surrogate

$$z_a^{\text{lin}}(\bar{h}) := z_a + c_a^\top \bar{h}.$$

701 This approximation is used as a first-order surrogate for branch opening, not as an exact global model
 702 of the transformer logits.

703 D.3.3 Top- K entry region (Remark)

704 This complements the informal branching argument in §5.1. Define the entry region

$$\mathcal{G}_{a,K} := \{\bar{h} \in \mathbb{R}^d : \#\{b \neq a : z_b^{\text{lin}}(\bar{h}) > z_a^{\text{lin}}(\bar{h})\} \leq K - 1\}.$$

705 If $\mathcal{G}_{a,K}$ contains a nonempty open set U , then by (S1) $D(U) > 0$, and $U \subseteq \mathcal{G}_{a,K}$ gives $\Pr_{\bar{h}}(\bar{h} \in$
 706 $\mathcal{G}_{a,K}) \geq D(U) > 0$. A useful sufficient condition is the existence of some $\bar{h}_0$ at which token a strictly
 707 outranks all but at most $K - 1$ other tokens in the affine surrogate; by continuity, a neighborhood of
 708 $\bar{h}_0$ is then contained in $\mathcal{G}_{a,K}$. This is a rank-changing event unreachable by the monotone temperature
 709 transform, but it does not by itself imply (M1).

710 D.3.4 Pass@ N ensemble bound (Remark)

711 The bound $\Pr(\text{Pass@}N \text{ on } x) \geq 1 - (1 - p_{\min}(x))^N$ in the main text is a standard independent-
 712 Bernoulli union argument. Let A_n be the event that the n -th run reaches a correct trajectory on x ;
 713 under (M1) $\Pr(A_n) \geq p_{\min}(x)$, and under (M2)

$$\Pr\left(\bigcap_{n=1}^N A_n^c\right) \leq (1 - p_{\min}(x))^N, \quad \Pr\left(\bigcup_{n=1}^N A_n\right) \geq 1 - (1 - p_{\min}(x))^N \geq 1 - e^{-N p_{\min}(x)}.$$

714 This is a generic fact, not specific to RSP; RSP's role is to make branch opening possible and
 715 to support (M2) through independent resampling, while whether a branch satisfies the task-side
 716 correctness condition (M1) remains task-dependent.

E Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 5 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

E.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,m)}$ denote the post-softmax attention weight at layer ℓ , head index m (using m to avoid collision with the vocab token a of §5.1 and the RSP matrix $\mathbf{H}$), from query position i to key position j , satisfying $\sum_j \alpha_{i,j}^{(\ell,m)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let N_{head} be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\text{PTM}_Q[\ell, i] = \frac{1}{|Q| N_{\text{head}}} \sum_{m=1}^{N_{\text{head}}} \sum_{j \in Q} \alpha_{i,j}^{(\ell,m)}, \quad \text{PTM}_R[\ell, i] = \frac{1}{|R| N_{\text{head}}} \sum_{m=1}^{N_{\text{head}}} \sum_{j \in R} \alpha_{i,j}^{(\ell,m)}. \quad (7)$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

E.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_{\bullet}^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \text{PTM}_{\bullet}[\ell, i]. \quad (8)$$

Figure 5 reports the sample average of Eq. (8) over the 500 valid samples.

E.3 Excluding the `\boxed{\dots}` Span

The reasoning position range terminates strictly before the final `\boxed{\dots}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{\dots}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

E.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. This choice is motivated by a standard late-layer effect rather than by any model-specific tuning decision.

Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

Excluding the final two layers keeps the heatmap focused on reasoning-phase dynamics instead of readout-phase dynamics. This exclusion is not load-bearing for the main claim: the sequence-direction attenuation emphasized in the main text is also visible without it, and Theorem 1 concerns sequence-direction attenuation rather than layer-wise monotonicity.

757 E.5 Additional Suffix Heatmaps

758 For the late-layer reasons discussed above, the pattern in Figure 7 does not generalize uniformly
 759 across every cell; nevertheless, the overall trend is consistent across both models: question-token
 760 attention varies broadly across layers, whereas RSP attention shows sequence-wise decay consistent
 761 with Theorem 1 alongside an empirical layer-wise attenuation that the theorem itself does not predict.

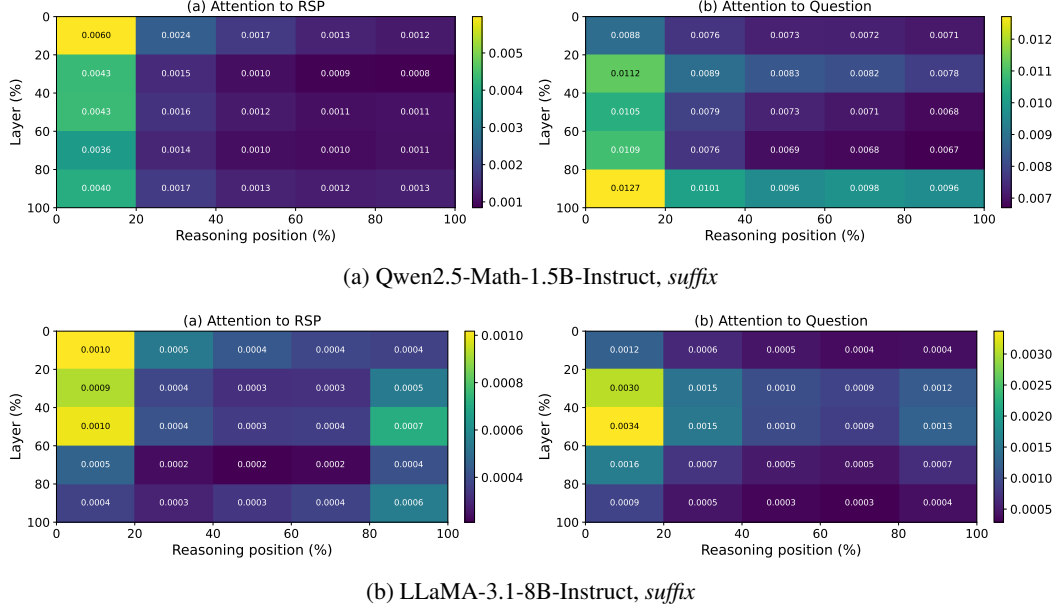


Figure 7: Per-token RSP attention mass under *suffix* injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 5: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

762 F DAPO Implementation

763 This appendix complements Section 4 with the DAPO loss modifications relative to GRPO, the
 764 DAPO + RSP implementation, and full per-step results.

765 F.1 From GRPO to DAPO

766 The vanilla GRPO objective [Shao et al., 2024] for a group of G rollouts $\{y_i\}_{i=1}^G$ (each $y_i =$
 767 $(y_{i,1}, \dots, y_{i,|y_i|})$ a token sequence; we use y rather than o throughout this appendix to avoid collision
 768 with the head output $\mathbf{o}_i^{(\ell)}$ of §5.1) with rewards $\{R_i\}_{i=1}^G$ is

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \left(\min(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon) \hat{A}_{i,t}) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \right) \right], \quad (9)$$

769 with importance ratio $r_{i,t} = \pi_\theta(y_{i,t} | q, y_{i,<t}) / \pi_{\theta_{\text{old}}}(y_{i,t} | q, y_{i,<t})$, group-relative advantage $\hat{A}_{i,t} =$
 770 $(R_i - \text{mean}(\{R_i\})) / \text{std}(\{R_i\})$, clipping range ε , and KL penalty βD_{KL} against the reference π_{ref} .

771 We build on the SimpleRL-Zoo [Zeng et al., 2025] training pipeline and implement DAPO on
 772 top of VeRL [Sheng et al., 2025]. DAPO modifies Eq. (9) in four ways: (i) token-level loss
 773 aggregation $1 / \sum_i |y_i|$ instead of $1 / |y_i|$, (ii) asymmetric clipping with $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}}) = (0.2, 0.28)$, (iii)
 774 an implementation-side negative-advantage dual-clipping safeguard $\max(L_{\text{clipped}}, c\hat{A})$ with $c = 10$
 775 (active only when $\hat{A} < 0$; omitted from Eq. (10) below for readability), and (iv) KL terms removed.
 776 The resulting DAPO objective is

$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_i |y_i|} \sum_{i,t} \min \left(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon_{\text{low}}, 1+\varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \right]. \quad (10)$$

777 F.2 DAPO + RSP Implementation

778 For DAPO + RSP, we additionally inject a per-rollout RSP $\mathbf{H}_i \in \mathbb{R}^{L \times d}$ ($L = 20$) into each rollout.
 779 Each $\mathbf{H}_i$ is generated deterministically from a per-step rollout seed and injected via a forward hook;
 780 it is not a trainable parameter, so the learning signal is the policy adapting to arbitrary $\mathbf{H}_i$ rather than
 781 a learned prompt. Including $\mathbf{H}_i$ in the log-probabilities at both rollout and update time keeps the
 782 update on-policy with respect to the RSP-conditioned distribution (avoiding off-policy mismatch).
 783 With $G = 8$ rollouts per prompt, each rollout receives an independently drawn $\mathbf{H}_i$.

784 F.3 Data, Batch, and Hyperparameters

Table 10: DAPO / DAPO + RSP training setup on Qwen2.5-Math-7B.

Field	Value
Train data	MATH Level 3–5 subset (<code>simplelr_math_35</code> , ~8,500 prompts)
Prompt template	<code>qwen-boxed</code>
Train batch size	1024
PPO mini-batch size	256 (4 PPO updates per rollout)
Rollouts per prompt G	8
Max prompt / response length	2,028 / 2,048
Rollout sampling	temperature 1.0, top- p 1.0, top- k 50
$(\epsilon_{\text{low}}, \epsilon_{\text{high}})$	(0.2, 0.28)
Dual clip c	10.0
Loss aggregation	token-mean ($1 / \sum_i y_i $)
KL terms	disabled
Entropy coefficient	0
Optimizer	AdamW, learning rate 5×10^{-7} (constant), no warmup
Total training	~10 epochs, ≥ 80 optimization steps
Save / eval frequency	every 10 steps
RSP (DAPO + RSP only)	<i>suffix</i> injection, $L = 20$, freshly drawn per rollout
Hardware	4× NVIDIA B200 GPUs
Wall-clock training time	~12 hours per run

785 F.4 Evaluation Protocol

786 Each saved checkpoint is converted to HuggingFace format and evaluated using the SimpleRL-
787 Zoo `eval_math_nodes.sh` pipeline, which selects sampling parameters per benchmark to reduce
788 variance on the smaller competition sets:

Table 11: Per-benchmark evaluation protocol used by SimpleRL-Zoo `sh/eval.sh`.

Benchmark	# Problems	Temperature	n_{sampling}	Metric
AIME 2024	30	1.0	32	Avg@32
AMC 2023	40	1.0	32	Avg@32
MATH-500	500	0.0	1	accuracy (mean@1)
GSM8K	1,319	0.0	1	accuracy
OlympiadBench	675	0.0	1	accuracy
Minerva Math	272	0.0	1	accuracy
GaoKao 2023-EN	385	0.0	1	accuracy

789 The maximum number of generated tokens is 16,000 across all benchmarks. The reported average is
790 the unweighted mean over the five benchmarks (GSM8K, MATH-500, College Math, Minerva Math,
791 AIME 2024).

Table 12: Five-benchmark average accuracy (%) at every checkpoint, computed over GSM8K, MATH-500, College Math, Minerva Math, and AIME 2024. **Bold** marks each method’s peak.

Step	DAPO	DAPO + RSP
0	30.06	30.06
10	46.40	46.22
20	49.02	49.54
30	49.58	50.72
40	50.62	52.06
50	50.88	52.40
60	51.84	52.96
70	52.52	53.86
80	53.20	54.04
90	52.52	54.36
100	50.54	53.64

793 G Broader Impacts

794 This work provides empirical and theoretical analysis of how random embedding injection affects
795 the reasoning behavior of large language models. The contributions are primarily foundational,
796 with potential downstream implications for reinforcement learning-based post-training of reasoning
797 models such as GRPO.

798 **Positive impacts.** RSP-induced trajectory diversity could improve the sample efficiency of RL-
799 based LLM post-training by providing richer learning signals within group-relative advantage estima-
800 tion. This has the potential to reduce compute costs associated with alignment and reasoning-oriented
801 fine-tuning, making such methods more accessible to research groups with limited resources.

802 **Potential concerns.** Methods that expand the effective output support of LLMs may increase the
803 reachability of low-probability tokens, which includes both desirable alternative reasoning paths
804 and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems
805 should combine it with existing safety filtering and alignment mechanisms rather than treating it as a
806 standalone diversity source.

807 **Limitations on impact scope.** Since this work focuses on analysis rather than deploying a new
808 model or dataset, the direct societal impact is limited. The broader implications described above are
809 contingent on future work integrating RSP into applied training pipelines.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and §1 state four scoped claims, each tied to a section: (i) RSP’s contribution at each attention layer is captured by a single scalar with a fixed-gap upper-envelope cache-growth bound (Theorem 1, §5.2); (ii) under a local affine surrogate, isotropic re-sampling reaches top- K entry regions unattainable by rank-preserving temperature sampling (§5.1); (iii) RSP raises early-token entropy and stabilizes later, with Pass@ N gains under independent re-sampling (§3); and (iv) the diversity transfers to RL training when combined with DAPO (§4). We do not claim a theorem that directly explains the format-mismatch recovery (Table 1); the wrong-attractor hypothesis there is explicitly informal.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Limitations are noted in several places rather than in a dedicated section: (a) the multi-path Pass@ N argument requires the task-side assumption $p_{\min}(x) > 0$, flagged in §5.2; (b) the +18–29 pp recovery on the format-mismatch rows of Table 1 is empirically observed but not formally derived from any theorem in §5, with the EOS-derailment hypothesis stated informally in §3.3 and its quantitative verification left for future work; (c) Table 1 reports per-position accuracy across {Prefix, Suffix}, with per-position variance large and several cases below baseline, so position is treated as a hyperparameter, and the RSP length L is selected per model from {10, 15, 20} on the evaluation set with held-out validation left for future work (§6); (d) Theorem 1 guarantees only sequence-axis attenuation, while the observed layer-axis attenuation is an empirical phenomenon from deep-layer attention-logit gap sharpening that lies outside the theorem’s envelope (§6); (e) the results are confined to RoPE-based decoders and math reasoning, with generalization to other position encodings and domains left for future work (§6). The future-work items in §6, namely generalization beyond RoPE/math reasoning, deep-layer sharpening, and held-out hyperparameter selection, correspond respectively to (e), (d), and (c).

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Theorem 1 (cache-growth attenuation) carries an inline proof in §5.2. Proposition 1 is stated in §5.1 with its proof in Appendix C; the supporting top- K entry and Pass@ N ensemble arguments around it are presented as informal discussion in §5.1 with formal write-ups in the appendix. Each statement explicitly lists its assumptions: finite logits and $n, L \geq 1$ for Theorem 1; finite second-moment budget for Proposition 1; and the locally affine surrogate plus $p_{\min}(x) > 0$ task-side condition together with mutual independence of the N runs for the Pass@ N argument.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: §3.2 and Appendix A specify benchmarks (MATH-500, GSM8K, AIME 2024 in the main table; further sets used for the DAPO study), models (LLaMA-3.1-8B-Instruct and Qwen2.5-Math 1.5B/7B variants), the RSP injection grid (Prefix/Infix/Suffix; $L \in \{10, 15, 20\}$), greedy decoding, and the answer extraction pipeline. Appendix F provides the full DAPO + RSP training recipe (batch of 1,024 prompts, $G = 8$ rollouts per

prompt at $\tau = 1.0$, four PPO mini-batch updates of 256 prompts each, 100 training steps, suffix RSP $L = 20$ during rollouts only).

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: All datasets and base models used in the paper are publicly available and cited (MATH-500, GSM8K, AIME 2024, AMC 2023, OlympiadBench, Minerva Math, GaoKao 2023-EN, College Math; Qwen2.5-Math 1.5B/7B and LLaMA-3.1-8B-Instruct). RSP itself is described to a level of detail sufficient for re-implementation (Eq. (1) and §3.2). However, our RSP injection harness and DAPO + RSP training pipeline are not yet released; we will release an anonymized implementation alongside the camera-ready submission.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: §3.2 and Appendix A list all benchmarks, models, RSP injection settings (positions, lengths, per-row L choice in Appendix Table 7), maximum generation length per benchmark, and decoding strategy. Appendix F provides the full DAPO + RSP training settings, including optimizer, batch composition, mini-batch updates, asymmetric clipping range $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}}) = (0.2, 0.28)$, dual clipping safeguard, and RSP injection protocol during rollouts.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Pass@ N experiments use $N = 16$ samples per problem (§3.5), Table 3 reports mean $\pm$ standard deviation over 16 RSP seeds, and AIME 2024 is scored by Avg@32 to reduce variance. However, Table 1 (Table 1) reports single-run accuracies without bootstrap confidence intervals or paired tests, and the DAPO + RSP training in §4 uses a single training seed. Multi-seed evaluation, particularly on small competition sets (AIME 2024, AMC 2023) where variance is highest, is left for future work.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Inference experiments (Tables 1, 2; Figure 2) run on a single NVIDIA A6000 40GB GPU with greedy decoding (Appendix A). DAPO + RSP training (§4) runs on 4 $\times$ NVIDIA B200 GPUs for approximately 12 hours per run (Appendix F, hardware row of the configuration table).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses publicly available math reasoning benchmarks and pretrained language models, involves no human subjects or personal data, and does not introduce a release that bypasses standard model-safety considerations. We have reviewed the NeurIPS Code of Ethics and identify no conflicts.

10. Broader impacts

914 Question: Does the paper discuss both potential positive societal impacts and negative
915 societal impacts of the work performed?

916 Answer: [Yes]

917 Justification: Appendix G discusses both directions: positive impact via more sample-
918 efficient RL post-training of reasoning LLMs, and a potential downside that broadening the
919 effective output support also increases reachability of undesirable trajectories, which we
920 argue should keep RSP coupled with existing safety filtering rather than replacing it.

921 **11. Safeguards**

922 Question: Does the paper describe safeguards that have been put in place for responsible
923 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
924 image generators, or scraped datasets)?

925 Answer: [N/A]

926 Justification: The paper does not release new pretrained models, generative systems, or
927 scraped datasets. RSP is a generation-time perturbation applied to existing pretrained models
928 and introduces no separate artifact requiring safeguards.

929 **12. Licenses for existing assets**

930 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
931 the paper, properly credited and are the license and terms of use explicitly mentioned and
932 properly respected?

933 Answer: [Yes]

934 Justification: All datasets (MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021],
935 AIME 2024, AMC 2023, OlympiadBench [He et al., 2024], Minerva Math [Lewkowycz
936 et al., 2022], GaoKao 2023-EN, College Math) and pretrained models (Qwen2.5-Math
937 [Yang et al., 2024], LLaMA-3.1 [Grattafiori et al., 2024]) are publicly available and cited
938 at first use. The training pipeline builds on SimpleRL-Zoo [Zeng et al., 2025] and VeRL
939 [Sheng et al., 2025], also cited. Each asset is used under its original publicly stated terms.

940 **13. New assets**

941 Question: Are new assets introduced in the paper well documented and is the documentation
942 provided alongside the assets?

943 Answer: [N/A]

944 Justification: The paper does not release new datasets or pretrained models. The implemen-
945 tation code planned for camera-ready release (item 5) will be accompanied by configuration
946 files and a README; no derived dataset or model checkpoint is part of the submission.

947 **14. Crowdsourcing and research with human subjects**

948 Question: For crowdsourcing experiments and research with human subjects, does the paper
949 include the full text of instructions given to participants and screenshots, if applicable, as
950 well as details about compensation (if any)?

951 Answer: [N/A]

952 Justification: The paper involves no crowdsourcing and no research with human subjects.
953 All evaluations use static, publicly available math reasoning benchmarks.

954 **15. Institutional review board (IRB) approvals or equivalent for research with human
955 subjects**

956 Question: Does the paper describe potential risks incurred by study participants, whether
957 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
958 approvals (or an equivalent approval/review based on the requirements of your country or
959 institution) were obtained?

960 Answer: [N/A]

961 Justification: As noted in item 14, the paper involves no human subjects, so IRB review is
962 not applicable.

963 **16. Declaration of LLM usage**

964 Question: Does the paper describe the usage of LLMs if it is an important, original, or
965 non-standard component of the core methods in this research? Note that if the LLM is used
966 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
967 scientific rigor, or originality of the research, declaration is not required.

968 Answer: [N/A]

969 Justification: Pretrained LLMs (Qwen2.5-Math, LLaMA-3.1) appear in the paper as the
970 experimental subject of analysis, not as a non-standard component of the methods themselves.
971 The methods (RSP definition, attention decomposition, DAPO + RSP training) do not rely
972 on auxiliary LLMs in any non-standard or originality-affecting role.